

ALEXANDER IMMANUEL GOERKE ZAMORA

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SUMMARY

Industrial Electronics and Automation Engineering student at UPV, completing Erasmus+ at DTU. Background in robotics, embedded control, and biosignal processing — with a completed Bachelor's thesis that delivered a prototype to an active PhD project at DTU and six months of paid engineering work in medical technology R&D.

EXPERIENCE

CEBREO MEDICAL A/S

Copenhagen, Denmark

Student Worker

Jan 2026 – Jun 2026

- Developed signal processing algorithms and analysis workflows for physiological biosignals (EEG, accelerometer, PPG) in a medical technology environment.
- Built data transformation pipelines and visualization tools for clinical biosignal inspection and signal quality assessment.
- Worked within a multidisciplinary team (software, physiology, product engineering) on real-world sensor data challenges including noise and signal quality.

PRIVATE TUTORING

Valencia, Spain

Maths, Physics & Chemistry Tutor

2022 – Present

- One-to-one instruction for secondary school students; designed study plans and applied structured problem-solving to improve academic outcomes.
- Recommended by school teachers following strong academic results; developed clear communication and mentoring skills.

EDUCATION

UNIVERSITAT POLITÈCNICA DE VALÈNCIA (UPV)

Valencia, Spain

B.Sc. Industrial Electronics and Automation Engineering

2022 – Present

- **Thesis (TFG):** Control design of a differential robotic arm for agricultural weed-intervention; prototype contributed to a 5-year PhD project at DTU.

TECHNICAL UNIVERSITY OF DENMARK (DTU)

Lyngby, Denmark

Erasmus+ Exchange

2025 – Present

- Courses: Robotics, Digital Control, Deep Learning, Business Analytics, Business Studies.

TECHNICAL SKILLS

- **Engineering:** Robotics, Digital Control, Signal Processing, Embedded Systems, Circuit Design, IK/FK Kinematics, Sensor Fusion, Mechatronics
- **Programming:** Python, C/C++, MATLAB, Arduino, HTML/CSS
- **Design & Simulation:** SolidWorks, AutoCAD, MATLAB/Simulink, Proteus, OrCAD
- **Tools:** Git, VSCode, LaTeX

PROJECTS

BACHELOR THESIS – DIFFERENTIAL ROBOT ARM

DTU / UPV, 2025–2026

Control Design for Agricultural Weed Intervention

- Designed a differential robotic arm prototype contributing to an active 5-year PhD research programme at DTU focused on agricultural weed-intervention.
- Derived inverse/forward kinematics, built a full Python simulation suite, and implemented a real-robot C++ controller on Teensy 4.1 with Dynamixel servos.
- Validated and compared feedforward, smooth bounded feedback, and full PID control strategies through quantitative real-robot experiments.

VISION-GUIDED 4-DOF ROBOT

DTU Robotics, 2025

- Built a robot to detect objects, convert pixel coordinates to world space, and reach them via IK; implemented HSV segmentation, camera calibration, and Dynamixel servo control.

PHYSICS-INFORMED GNN FOR MOLECULAR PROPERTY PREDICTION

DTU Deep Learning, 2025

- Trained a PaiNN graph neural network on QM9 and MD17 datasets for molecular property prediction.

CERTIFICATIONS & AWARDS

- **ISO 21500 & ISO 21502** – Project, Programme and Portfolio Management (Danish Standards, Jan 2026)
- Advanced English (C1) | German (A2) | Web Programming (HTML/CSS/JS) | Energy Systems (CFP-UPV, 2024)
- Science Prize – Iale International School

Languages: Spanish (native) | English (C1) | German (A2)